Generalized Fiducial Inference for Hybrid Discrete–Continuous Count Data

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Full Paper

Introduction

Motivation: Tracking data are summarized as a quantile cube (Fig 1): time in each of 100 velocity × acceleration × angle bins, from under 100 to thousands of deciseconds per bin.

Goal: Uncertainty for hundreds of parameters, with no prior.

Approach: Hybrid generalized fiducial inference: exact **negative binomial (NB)** where the NB shape is small, a Normal approximation where it is large.

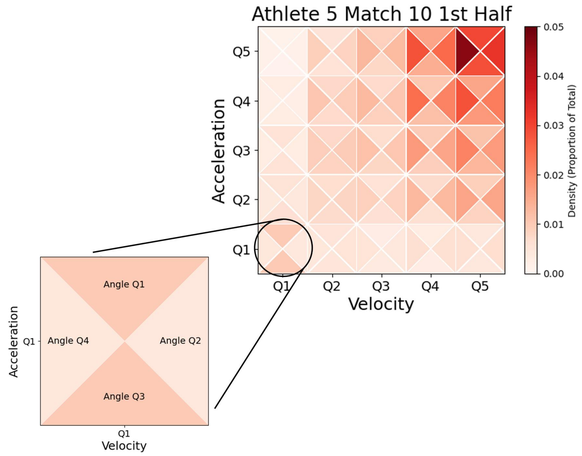


Fig 1. An example quantile cube (Thomas and Hannig, 2026): 3-D summary of an athlete's movement distribution for half of a match, shaded by share of time.

Data

17 professional athletes	**216** athlete–match sessions
100 movement bins (5×5×4)	**500** model parameters

- 10 Hz GPS from one professional women's soccer season
- Exposure: time in prescribed activity (match, warm-up and cool-down)
- Covariates: age (z-score), position (vs. defender)

Methods

Exact NB or Normal approximation? Decided by the estimated NB shape $\hat{r}_{ij}$

Exact NB | **Normal approximation**

$\hat{r}_{ij}^{L} < \omega$ **cutoff $\omega = 10$** $\hat{r}_{ij}^{L} \geq \omega$

$$r(\theta|y) \propto L(y|\theta) \times J(\theta, y_C)$$

Generalized Fiducial Density ∝ likelihood × Jacobian

the likelihood uses every observation
the Jacobian uses only the normal-approximated ones, $\boldsymbol{y}_C$, and plays the role of the prior

- Block-diagonal Jacobian: each of 100 bins fit separately
- Bernstein–von Mises Theorem: matches flat-prior Bayes as n grows

NB shape $r_{ij} = \mu_{ij}\nu_j$ (mean × dispersion): the larger it is, the closer count i in bin j is to Normal.

Estimate: $\hat{r}_{ij} = \hat{\mu}_{ij}\hat{\nu}_j$, with $\hat{\mu}_{ij}$ the bin's average count per minute × session i's minutes and $\hat{\nu}_j$ a moment estimate.

- $\hat{r}_{ij}^{L}$ is a 95% lower bound on $\hat{r}_{ij}$ that allows for estimation error, by resampling athletes.

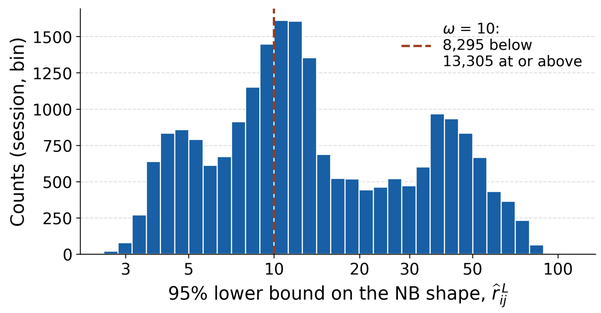


Fig 2. 95% lower bounds $\hat{r}_{ij}^{L}$, one per count; 61.6% reach $\omega = 10$ and use the Normal approximation.

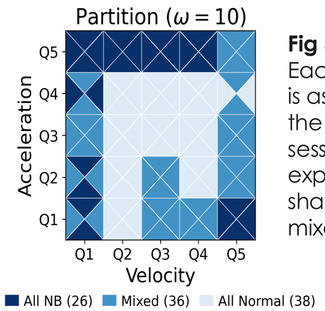


Fig 3. Partition at ω = 10. Each count (session x bin) is assigned separately, not the whole bin: longer sessions have larger expected counts and shapes, so one bin can be mixed over the dataset.

Simulation Results

Simulated NB regressions (intercept, continuous covariate, 3-level group, exposure):

- 3 regimes by NB shape: sparse (all exact NB), borderline (mixed), heavy (all Normal)
- n = 50 to 500
- 2,000 replications each

94.6% average coverage of 95% intervals (93–96%)

No prior uncertainty

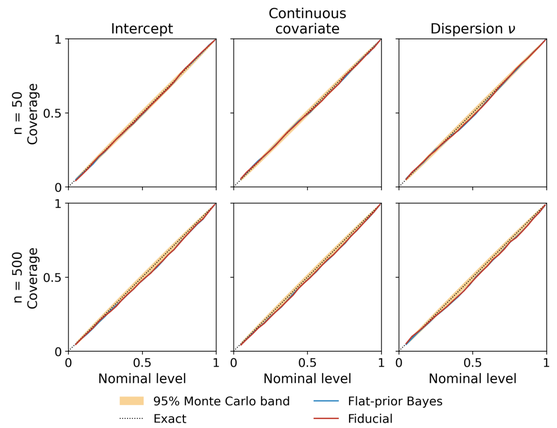


Fig 4. Coverage at every nominal level (borderline regime): fiducial (red) on the diagonal; flat-prior Bayes (blue) for reference.

Partition by shape, not by mean

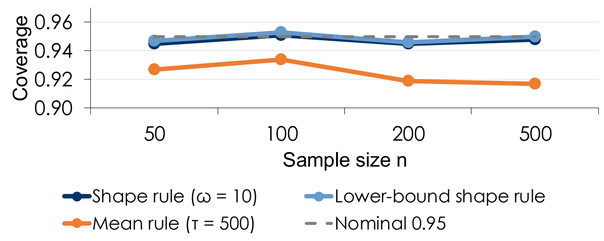
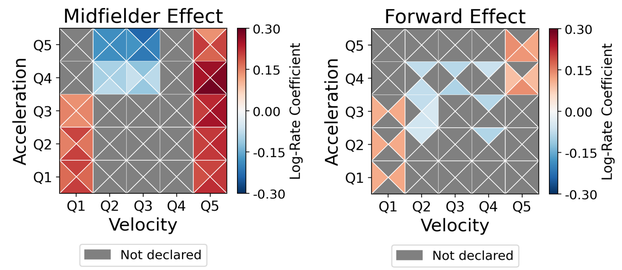


Fig 5. Average coverage of 95% intervals, borderline regime, ν varying; mean rule: Normal if the mean ≥ τ = 500. Monte Carlo SE ≈ 0.007.

Partitioning by the realized counts fails: intercept coverage drops from 0.92 to 0.71 as n grows.

Why $\omega = 10$: the Normal approximation holds once the NB shape is about 10, and coverage deteriorates for shapes near 2.7.

Application Results



Position effects reach $|\beta| = 0.29$
age effects stay below 0.09 per 2.1 years (1 SD) of age

Caution: an athlete-level permutation check shows these patterns are not distinguishable from label noise (17 athletes, not 216 independent units).

Fig 6. Fiducial mean effect per bin (log-rate). Colored bins: sign declared at Benjamini–Hochberg (BH) level 0.05 under the working model; triangles = angle sectors; gray = not declared.

Conclusions & Future Directions

- Near-nominal coverage (94.6% on average) in discrete, mixed and Normal regimes
- Partition by estimated shape, not by mean or realized counts
- Position effects cluster at velocity extremes; age effects follow an acceleration gradient (descriptive only)

Future: athlete-level random effects, fixed-sum constraint, multi-season data

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